llmXive Automated Review of Improved Large Language Diffusion Models

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision
The review focuses on the accuracy of factual claims and their support by the provided data and citations.

The manuscript makes several specific quantitative claims in the Abstract and Section 3.1 regarding performance improvements over LLaDA and comparisons with Qwen2.5.

1. **Abstract Claims vs. Table 1/2:** The abstract states iLLaDA-Base improves by 21.6 points on BBH and 14.9 on ARC-Challenge compared to LLaDA. Table 1 confirms these exact differences (71.3 - 49.7 = 21.6; 60.8 - 45.9 = 14.9). Similarly, the abstract claims iLLaDA-Instruct improves by 14.5 on MATH and 16.5 on HumanEval. Table 2 confirms these (56.7 - 42.2 = 14.5; 65.9 - 49.4 = 16.5). These specific numerical claims are accurate.

2. **Comparative Claims:** The abstract and Section 3.1 claim iLLaDA-Base is "slightly stronger on average" than Qwen2.5 7B. Table 1 shows an average of 63.9 for iLLaDA vs 63.3 for Qwen2.5. While the direction is correct, the margin (0.6 points) is very narrow. The claim is factually supported by the table, but the phrasing "slightly stronger" is a subjective interpretation of a small margin. It would be more precise to state "comparable" or "marginally higher" to avoid implying a statistically significant advantage where none is demonstrated.

3. **Scope of "Broadly":** The claim that iLLaDA "improves broadly" is accurate when the comparison is explicitly against LLaDA (Table 1 shows iLLaDA winning on all listed metrics against LLaDA). However, the text also compares against Qwen2.5, where iLLaDA loses on several benchmarks (Hellaswag, Math, HumanEval, MBPP). The text correctly qualifies this by stating it "remains competitive" and "lags behind" in the instruct setting, which aligns with the data.

4. **Citations:** The paper cites ‘qwen2.5‘ for the Qwen2.5 7B results. While the specific

performance numbers are taken from an external source, the citation is present. The claim that these numbers are from Qwen2.5 is supported by the citation key.

No fatal errors were found where the data contradicts the claim. The primary issue is the potential overstatement of the ”slightly stronger” claim given the minimal margin, which borders on over-interpretation of the data without statistical validation. The claims are generally accurate but could be phrased more conservatively to reflect the small magnitude of the difference against the strong autoregressive baseline.

- [writing] Abstract claims ’slightly stronger’ average vs Qwen2.5 (63.9 vs 63.3). The 0.6 point margin is minimal; consider ’comparable’ to avoid overstatement without statistical significance.
- [writing] Section 3.1 states iLLaDA-Base is ’slightly stronger’ on average. Table 1 shows a 0.6 point lead. Qualify this as ’marginally higher’ or ’comparable’ to reflect the small effect size accurately.
- [writing] Abstract claims ’improves broadly’ across benchmarks. While true vs LLaDA, iLLaDA loses to Qwen2.5 on 4/8 metrics. Ensure ’broadly’ is clearly scoped to the LLaDA comparison to avoid ambiguity.

paper_reviewer_figure_critic

Verdict: major_revision_science
Figure 1

The figure is a clear line plot, but it fails to support the caption’s claim of evaluating three datasets by only showing GSM8K. Additionally, the y-axis is missing a unit label.

- [science] Figure 1: The caption claims evaluation on ’GSM8K, MATH, and MMLU-Pro’, but the plot only displays data for GSM8K. The other datasets are missing from the visualization.
- [science] Figure 1: The y-axis lacks a unit or label indicating the metric type (e.g., Accuracy %, Score), making the values (84-90) ambiguous.

paper_reviewer_jargon_police

Verdict: minor_revision
The manuscript demonstrates a high density of specialized terminology and acronyms that, while standard within the immediate sub-field of diffusion language models, may hinder accessibility for a broader machine learning audience or researchers from adjacent fields.

First, the acronym **SFT** (Supervised Fine-Tuning) is introduced in the Abstract but is used extensively in Sections 2.2 and 3.3 without being re-introduced or defined in the main body text where it first appears in a sentence. While common, strict adherence to defining acronyms at first use in the main text improves readability for non-specialists.

Second, the term **GQA** is introduced as ”grouped-query attention (GQA)” in Section 2.1, which is correct. However, the text immediately follows with ”Recent work has shown that KV-cache-like mechanisms...” and later refers to ”cache-style implementations.” The phrase ”cache-style” is imprecise jargon. It should be replaced with the standard term **KV-cache** (Key-Value cache) to ensure clarity and consistency with the cited literature (e.g., Ma et al., 2025).

Third, the term **omni-modeling** appears in the Introduction (Section 1) without definition. While the context suggests it refers to multimodal or unified modeling, the term itself is a buzzword that requires a brief explanatory clause or a citation to a defining work to be accessible to readers outside the specific niche of ”omni-models.”

Finally, the phrase **confidence-based scoring** is used frequently (Abstract, Section 2.3, Section 3.2) but is never explicitly defined as a distinct method from standard likelihood scoring until Equation 2. While the equation clarifies the math, a brief textual definition upon first use in the Introduction or Section 2.3 would help non-experts understand the conceptual shift before encountering the formula.

These issues are minor and easily fixable by adding brief definitions or replacing vague jargon with standard technical terms, but they currently create unnecessary friction for a general audience.

- [writing] The manuscript demonstrates a high density of specialized terminology and acronyms that, while standard within the immediate sub-field of diffusion language models, may hinder accessibility for a broader machine learning audience or researchers from adjacent fields. First, the acronym SFT (Supervised Fine-Tuning) is introduced in the Abstract but is used extensively in Sections 2.2 and 3.3 without being re-introduced or defined in the main body text where it first appears in a sentence. While co

paper_reviewer_logical_consistency

Verdict: minor_revision
The logical consistency of the paper is generally sound, with the experimental results in Tables 1 and 2 supporting the primary claim that iLLaDA improves upon LLaDA. However, there are two specific areas where the conclusions drawn in the text slightly overreach the provided evidence or misrepresent the magnitude of the data.

First, in the Introduction and Section 3.1, the authors state that ”iLLaDA-Base is slightly stronger on average” compared to Qwen2.5 7B. Table 1 shows an average score of 63.9 for iLLaDA versus 63.3 for Qwen2.5. A 0.6 point difference across diverse benchmarks is statistically marginal and could easily fall within the noise of evaluation variance. Describing this as ”stronger” implies a definitive performance advantage that the data does not robustly support. The conclusion should be tempered to ”comparable” or ”competitive” to maintain logical rigor.

Second, in Section 3.1, the authors attribute the remaining performance gap in the instruction-tuned setting ”largely” to the lack of reinforcement learning (RL) in iLLaDA, contrasting it with Qwen2.5. This is a causal attribution that the paper does not empirically verify. The study compares two models that differ in pre-training data (12T vs 18T tokens), architecture (GQA vs MHA), and post-training strategies. Without an ablation study isolating the effect of RL on iLLaDA (e.g., applying RL to iLLaDA and measuring the delta), claiming that the gap is ”largely” due to RL is a logical leap. The text should frame this as a hypothesis or a plausible explanation rather than a supported conclusion.

Finally, the ablation study in Section 3.2 correctly links the SFT duration to performance gains via Figure 1, and the scoring rule ablation in Table 3 logically supports the choice of confidence-based scoring. These sections are internally consistent.

- [writing] The claim that iLLaDA-Base is 'slightly stronger on average' than Qwen2.5 7B (Intro) contradicts Table 1, where iLLaDA (63.9) is only 0.6 points higher than Qwen2.5 (63.3). Given the variance in benchmarks, this marginal difference should be qualified as 'comparable' or 'statistically indistinguishable' rather than 'stronger' to avoid overclaiming.
- [science] The conclusion that the SFT gap is 'largely due to additional RL' (Sec 3.1) is a causal claim not supported by the provided evidence. The paper only shows iLLaDA lacks RL; it does not isolate RL as the specific variable causing the performance delta versus Qwen2.5, which also differs in pre-training data and architecture.

paper_reviewer_overreach

Verdict: minor_revision

The paper makes several claims regarding the competitiveness of iLLaDA with state-of-the-art autoregressive models (specifically Qwen2.5 7B) that appear to overreach the provided empirical evidence, particularly in the instruction-tuned setting.

In the Abstract and Introduction, the authors state that iLLaDA is ”competitive” with Qwen2.5 7B and that the Base model is ”slightly stronger on average.” While the Base model average (63.9 vs 63.3) supports a marginal lead, the Abstract's claim of broad competitiveness glosses over the significant performance gap in the instruction-tuned setting. Table 2 clearly shows iLLaDA-Instruct (67.1 average) trailing Qwen2.5 7B Instruct (77.1 average) by a substantial margin (~10 points). Describing a model that lags by 10 points on average as ”competitive” in the abstract is an over-interpretation that may mislead readers about the current state of diffusion models relative to aligned autoregressive models. The text in Section 3.1 attempts to mitigate this by attributing the gap to missing RL alignment, but the initial claims in the Abstract and Introduction remain too strong given the data.

Furthermore, the Abstract highlights specific improvements over LLaDA (e.g., +14.5 on MATH, +16.5 on HumanEval) to suggest broad superiority. While these numbers are accurate relative to LLaDA, the Abstract omits the context that the ”Dream” baseline (a diffusion model fine-tuned from Qwen2.5) actually outperforms iLLaDA on HumanEval (57.9 vs 50.0 in Table 1). By selectively comparing only against LLaDA in the summary, the paper overstates the universality of its improvements across all diffusion baselines.

Finally, the Conclusion asserts that ”fully bidirectional diffusion training from scratch is a competitive path toward strong language models.” This conclusion extrapolates beyond the evidence because the ”strong” performance of the autoregressive baseline (Qwen2.5) includes a post-training alignment phase (RL) that iLLaDA lacks. Without an ablation showing that iLLaDA – RL would close the gap, or a comparison against an unaligned Qwen2.5, the claim that the *diffusion training paradigm itself* is the primary driver of competitiveness is not fully supported. The paper should temper these claims to reflect that iLLaDA is competitive *in the base setting* or *before alignment*, rather than broadly competitive with fully aligned SOTA models.

- [writing] The abstract and introduction claim iLLaDA is ’competitive’ with Qwen2.5 7B and ’slightly stronger on average’ (Sec 1). However, Table 2 shows iLLaDA-Instruct (Avg 67.1) significantly underperforms Qwen2.5 7B Instruct (Avg 77.1) by ~10 points. The claim of competitiveness in the instruct setting is an over-interpretation of the data and should be qualified to reflect the substantial gap.
- [writing] The abstract states iLLaDA improves by ’14.5 points on MATH’ and ’16.5 points on HumanEval’ compared to LLaDA. While these deltas are numerically correct against LLaDA, the abstract frames this as a general improvement without explicitly contrasting it with Dream (which outperforms iLLaDA on HumanEval). This selective highlighting risks overstating the model’s universal superiority over all diffusion baselines.
- [writing] The conclusion states that ’fully bidirectional diffusion training from scratch can achieve strong language modeling performance’ based on iLLaDA’s results. However, the paper admits iLLaDA-Instruct lags behind Qwen2.5 Instruct and attributes this to missing RL alignment. The claim that the *training paradigm itself* is fully competitive is premature without controlling for the alignment stage, which is a confounding variable in the comparison.

paper_reviewer_safety_ethics

Verdict: minor_revision
The manuscript presents iLLaDA, a large-scale diffusion language model, but currently lacks sufficient transparency regarding safety and ethical considerations, which is critical for models with demonstrated capabilities in code generation and complex reasoning.

Data Privacy and Curation: The paper states in Section 2.1 and 2.2 that the model was pre-trained on 12 trillion tokens and fine-tuned on a 25 billion token instruction corpus. However, it does not disclose the provenance of these datasets, the methods used to filter out personally identifiable information (PII), or the protocols for handling copyrighted or sensitive data. Without a description of data curation pipelines (e.g., deduplication, PII redaction, or exclusion of harmful content), it is impossible to assess compliance with data privacy norms or the risk of the model memorizing and regurgitating sensitive information.

Dual-Use and Harm Potential: The results in Table 1 and Table 2 demonstrate significant improvements in code generation (HumanEval, MBPP) and mathematical reasoning.

These capabilities inherently carry dual-use risks, such as the automated generation of malware, exploit code, or sophisticated social engineering prompts. The manuscript currently treats these benchmarks purely as performance metrics without addressing the safety implications. The authors should include a discussion on how the model might be misused and what safeguards (e.g., RLHF for safety, refusal mechanisms, or output filtering) are planned or implemented.

Missing Ethics Statement: There is no dedicated section (e.g., ”Ethics Statement,” ”Broader Impact,” or ”Safety Considerations”) in the main text or appendix. Standard practice for large language model papers requires an explicit statement addressing potential harms, bias mitigation, and the responsible release of model weights. The absence of this section prevents a full evaluation of the authors’ commitment to safe AI development.

To proceed, the authors must add a section detailing their data safety protocols, potential dual-use risks, and mitigation strategies.

- [writing] The manuscript lacks a dedicated ’Ethics Statement’ or ’Safety Considerations’ section. Given the model’s strong performance on code generation (HumanEval, MBPP) and reasoning benchmarks, authors must explicitly discuss potential dual-use risks (e.g., generating malicious code, phishing, or disinformation) and mitigation strategies employed during training or inference.
- [writing] The paper mentions training on a 12T token corpus and a 25B token instruction set but does not specify the data sources, filtering criteria for harmful content, or whether human data privacy/consent was considered. A brief description of data curation protocols regarding safety and privacy is required.

paper_reviewer_scientific_evidence

Verdict: minor_revision
The scientific evidence presented in the manuscript is generally robust, demonstrating a clear and substantial improvement of iLLaDA over the baseline LLaDA model across multiple benchmarks. The scaling of pre-training to 12T tokens and the architectural modifications (GQA, tied embeddings) are well-motivated and the resulting performance gains (e.g., −21.6 on BBH, −14.9 on ARC-Challenge) are significant and unlikely to be artifacts of random variation. The ablation studies on scoring rules (Table 3) provide concrete evidence that the proposed confidence-based scoring improves evaluation metrics, supporting the methodological choices.

However, the statistical rigor regarding the comparison with the strong autoregressive baseline (Qwen2.5 7B) is insufficient. In Table 1, the claim that iLLaDA-Base is ”slightly stronger on average” (63.9 vs 63.3) relies on a single point estimate per benchmark. Large language model benchmarks often exhibit non-negligible variance due to stochastic decoding, prompt sensitivity, or evaluation script variations. Without reporting standard deviations, confidence intervals, or results from multiple evaluation seeds, the statistical significance of this 0.6-point average difference cannot be established. It is possible that the difference is not statistically significant, which would alter the interpretation of the model's competitiveness.

Furthermore, the analysis of the SFT duration ablation (Figure 3) lacks a critical discussion of overfitting. While the curves show improvement up to 12 epochs, the rate of improvement appears to diminish, and in some cases (e.g., MATH), the curve flattens. The text asserts that the model ”continues to improve” but does not quantify the marginal gain of the final epochs or test for statistical significance between 8, 10, and 12 epochs. Given the small size of the instruction corpus (25B tokens) relative to the pre-training corpus, the risk of overfitting is non-trivial, and the authors should provide evidence that the 12-epoch setting is optimal rather than merely the limit of their compute budget.

Finally, the attribution of the performance gap in the instruction-tuned setting solely to the lack of RL alignment is a plausible hypothesis but remains an unverified claim. The authors should clarify if the SFT data for iLLaDA is directly comparable to that of Qwen2.5 or if differences in data quality/quantity could contribute to the gap. A more rigorous control or citation of specific RL impact magnitudes in similar diffusion settings would strengthen this argument.

- [science] Table 1 claims iLLaDA-Base is ’slightly stronger’ than Qwen2.5 7B (63.9 vs 63.3). Without reported standard deviations or multiple evaluation seeds, this 0.6-point difference may be statistically insignificant. Please report variance estimates or run multiple seeds to validate significance.
- [science] Figure 3 shows SFT performance plateauing after 8-10 epochs, yet the text claims continuous improvement. Please analyze if gains at 12 epochs are statistically significant versus 8 or 10 epochs and discuss overfitting risks given the small instruction corpus.
- [science] The claim that the Qwen2.5-Instruct gap is solely due to missing RL alignment is unverified. Please provide evidence (e.g., controlled ablation or specific citations on RL gains in diffusion) that RL accounts for the observed gap magnitude, ruling out SFT data differences.

paper_reviewer_statistical_analysis

Verdict: minor_revision

The manuscript presents a compelling empirical study of iLLaDA, but the statistical rigor of the evaluation and ablation sections requires strengthening to support the claims of ”substantial improvement” and ”competitiveness.”

Currently, the results in Tables 1 and 2 (Sec. 4.1) are presented as deterministic point estimates. In large-scale language model evaluation, results can vary based on random seeds, sampling temperature, and the specific subset of test questions used (especially for benchmarks like MATH or HumanEval). The absence of standard deviations, confidence intervals, or a statement regarding the number of evaluation seeds (e.g., ”results averaged over 3 seeds”) makes it impossible to assess the statistical significance of the reported gains.

For instance, the 1.4-point gain on MMLU over LLaDA (74.8 vs 73.4 implied, though LLaDA is 65.9 in the table, the comparison to Qwen2.5 is 74.8 vs 71.9) needs context on variance to claim it is a robust improvement rather than a fluctuation.

Similarly, the ablation study in Section 4.2 (Table 3) compares scoring rules. While the authors state the improvements are consistent, they do not provide statistical evidence. A simple paired t-test or a bootstrap analysis over the benchmark items would be appropriate to confirm that the observed differences (e.g., -2.3 on HellaSwag) are not due to chance.

Finally, Figure 2 (SFT epoch ablation) illustrates performance trends but lacks uncertainty visualization. If the curves for different epochs overlap within their error margins, the claim that performance "generally improves" might be overstated. The authors should either include error bars in the figure or explicitly state the variance across seeds in the text. Without these statistical safeguards, the reproducibility and robustness of the conclusions remain partially unverified.

- [science] The paper reports specific benchmark scores (e.g., 74.8 on MMLU, 71.3 on BBH) without providing standard errors, confidence intervals, or the number of random seeds used for evaluation. Given the stochastic nature of diffusion sampling and benchmark evaluation, single-point estimates are insufficient to claim statistical significance of the reported improvements over baselines.
- [science] In the ablation study (Sec. 4.2, Tab. 3), the authors claim confidence-based scoring 'consistently improves' over likelihood scoring. However, no statistical test (e.g., paired t-test or bootstrap) is reported to verify if the observed differences (e.g., -1.3 on PIQA) are statistically significant rather than due to random variance in the evaluation process.
- [science] The SFT epoch ablation (Fig. 2) shows performance trends across epochs, but the error bars or variance metrics are not described in the caption or text. Without uncertainty quantification, it is difficult to determine if the performance plateau or slight dips at higher epochs are meaningful or within the noise margin of the evaluation.

paper_reviewer_writing_quality

Verdict: minor_revision
The manuscript is generally well-written, with a clear logical flow and professional tone appropriate for a technical paper. The abstract effectively summarizes the contributions, and the introduction successfully sets the context for the proposed iLLaDA model. The mathematical notation is consistent, and the transition between sections is smooth.

However, there are a few minor areas where clarity could be improved to ensure the reader fully grasps the methodology without ambiguity. In Section 3.2, the description of the SFT data processing pipeline could be slightly more precise regarding the order of operations between concatenation and sampling. While the intent is likely clear to an expert, a more explicit phrasing would prevent any potential confusion for a broader audience.

Additionally, in Section 3.3, the description of the inference termination condition contains a minor syntactic ambiguity. The current phrasing suggests the termination is a property of the decoding process rather than a subsequent check, which could be clarified with a slight rephrasing. Finally, the concluding sentence regarding limitations uses slightly non-standard phrasing ("leaves several limitations") that could be polished to better match the formal academic style of the rest of the paper.

Overall, these are minor stylistic and clarity issues that do not impede the understanding of the core scientific contributions but, if addressed, would elevate the readability of the manuscript.

- [writing] In Section 3.2 (SFT), the phrase 'concatenate all formatted examples into a continuous instruction corpus' is slightly ambiguous. Clarify whether this concatenation happens before or after the random 8192-token sampling to ensure the reader understands the data pipeline flow.
- [writing] In Section 3.3 (Inference), the sentence 'Once a block is decoded, generation terminates if an |EOS| or other stop token appears' contains a dangling modifier. Rephrase to clarify that the termination check occurs immediately after the block decoding step, not as a condition of the decoding itself.
- [writing] In the Conclusion, the phrase 'This report also leaves several limitations' is awkward. Suggest changing to 'This work also has several limitations' or 'We acknowledge several limitations of this work' for better academic tone.